

MATERNOVA API ECOSYSTEM

Recommended Third-Party Services for National Award Integration

Overview: To elevate the Maternova platform without incurring significant infrastructure costs, the following free-tier and open-source APIs have been selected. These services provide the necessary deep-tech intelligence, localized accessibility, and public health interoperability required for a national award-winning GovTech solution.

1. Anthropic Claude API

(Core AI Engine)

Free Tier Options

Description: A state-of-the-art large language model known for its exceptional reasoning, coding capabilities, and ability to generate natural, human-like responses. It features strong multimodal vision capabilities.

Maternova Use Case: Utilized for complex vision tasks, specifically Optical Character Recognition (OCR) to read and extract structured data from messy, handwritten clinical notes and localized medical reports submitted by ASHA workers or rural patients.

2. Bhashini API

(Vernacular Translation)

Free for Developers

Description: An initiative by the Government of India providing highly accurate translation, text-to-speech, and speech-to-text models across 22 Indian languages.

Maternova Use Case: Crucial for breaking the literacy and language barrier. It powers the voice-first interface, allowing ASHA workers and elderly users to interact with the platform in their native regional dialects rather than English.

3. Infermedica API

(Medical Triage & Symptoms)

Free Developer Portal

Description: A clinically validated symptom-checking and medical triage engine. It uses structured diagnostic pipelines rather than generative text to evaluate patient symptoms safely.

Maternova Use Case: Replaces general LLM "hallucinations" for medical advice. It is used to securely process maternal and geriatric symptom logs to accurately flag risk levels (e.g., signs of preeclampsia) for ASHA workers.

4. DHIS2 API

(Community Health Integration)

Open Source / Free

Description: District Health Information Software 2 is the world's largest health management information system platform, widely used by governments to collect and analyze community health data.

Maternova Use Case: Provides the structural backbone for the platform's GovTech integration. It allows the offline-collected data from ASHA workers to seamlessly sync and map to standardized national health registries when internet connectivity is restored.